

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 2b: Theory of Probability

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College of Engineering

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Outline

- ① Theory of Probability
- ② Addition Rule
- ③ Counting methods
- ④ Outlook
- ⑤ Appendix

Recap from Lecture 2a

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- De Morgan's Rules are useful for expressing complements of unions or of intersections:

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

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- Perform data input, set operations, permutations and combinations in MATLAB

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$$P(BSc) = \frac{2}{3+2} = \frac{2}{5} = 0.40$$

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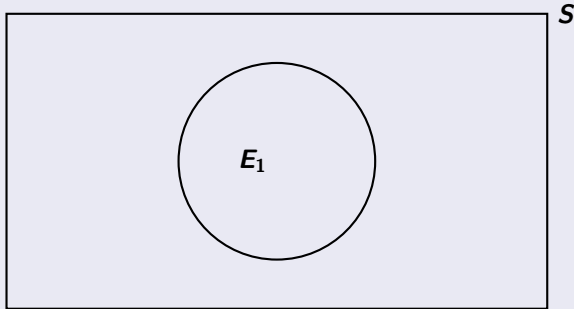
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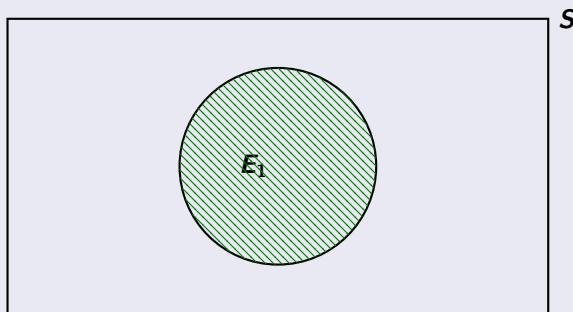
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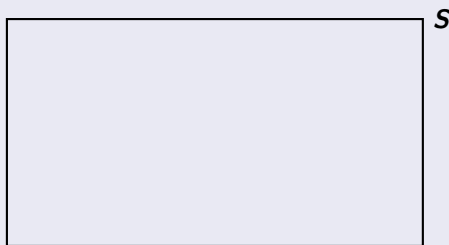
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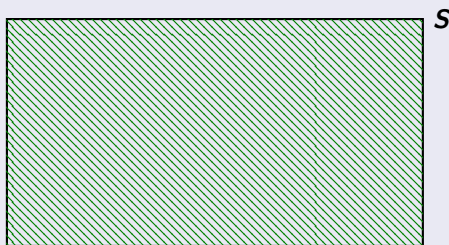
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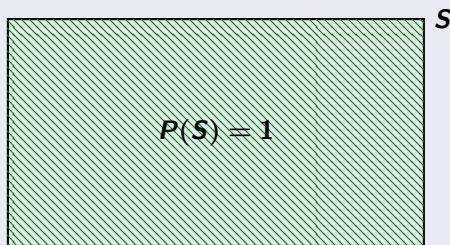
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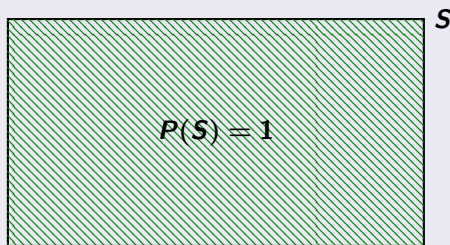
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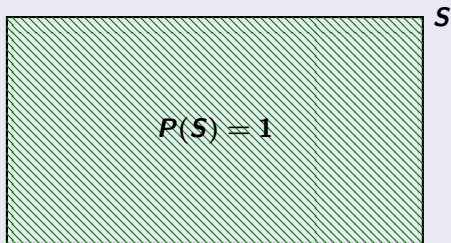
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In other words, the probability of the sample space is unity (or 1).

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If E_1 and E_2 are mutually exclusive events with probabilities 0.2 and 0.3, respectively. Find $P(E_1 \cup E_2)$.

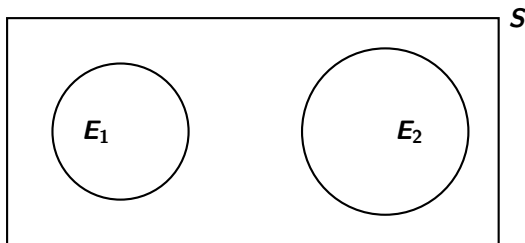
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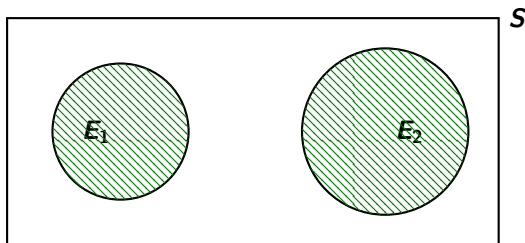
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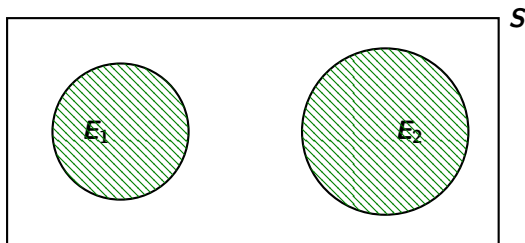
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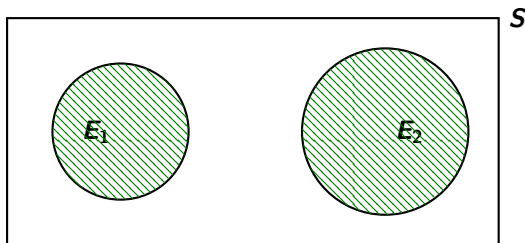
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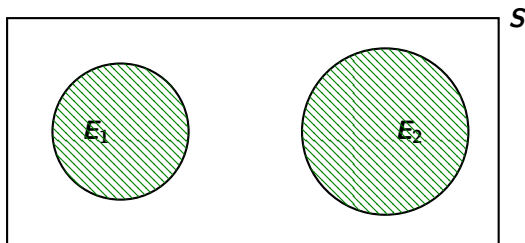
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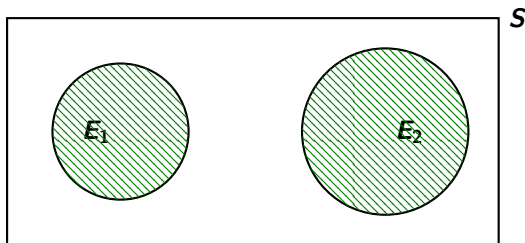
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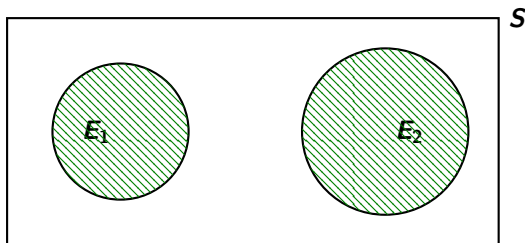
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$$0 \leq P(E) \leq 1 \quad (5)$$

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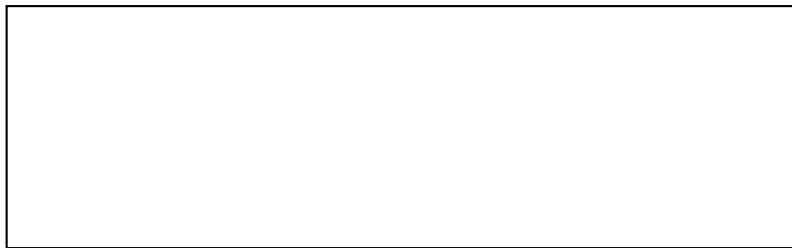
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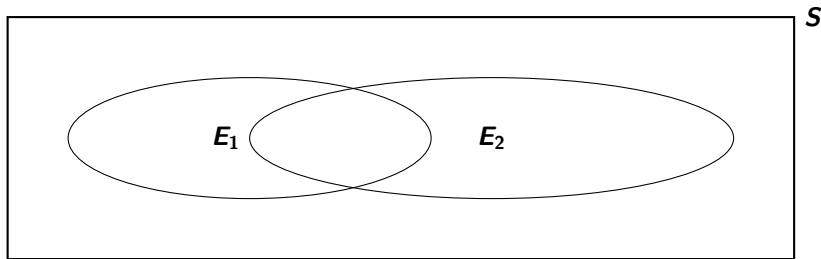
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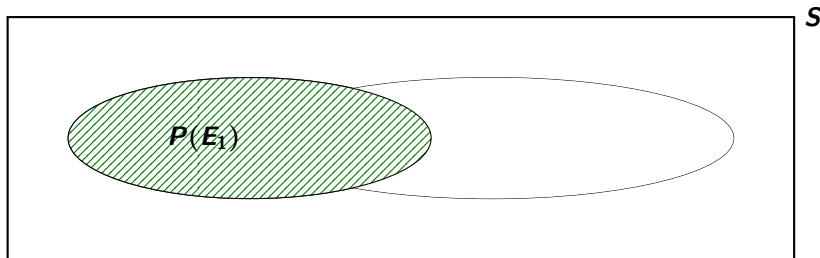
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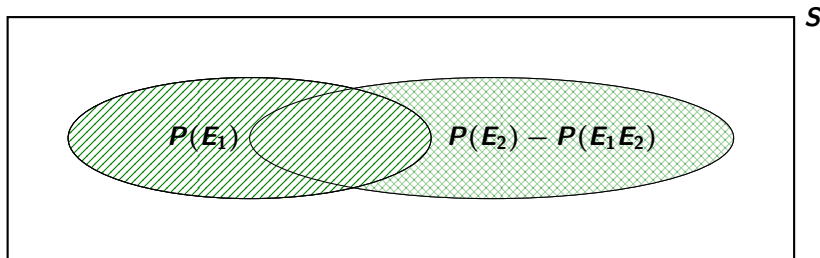
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which yields Axiom 3.

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3	14	14/60
4	3	3/60
5	2	2/60
6	1	1/60
7	0	0
8	0	0

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Given the following definitions:

$E_1 = > 2$ vehicles waiting for left turns

$E_2 = \leq 4$ vehicles waiting for left turns

Example 2: Left-turn lane design

60 observations of the number of vehicles waiting for left turns at an intersection are shown in the table.

No. of Vehicles Waiting	No. of Observations	Relative Frequency
0	4	4/60
1	16	16/60
2	20	20/60
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Find the following probabilities:

- (a) $P(E_1)$
- (b) $P(E_2)$
- (c) $P(E_1 E_2)$
- (d) $P(E_1 \cup E_2)$

Example 2: Left-turn lane design (cont.)

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Let X be the number of vehicles.

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How would you characterize E_1 and E_2 ?

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Collectively exhaustive events

Fundamental principle of counting

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Given $1, 2, \dots, k$ operations are to be performed, and $n_1, n_2, \dots, n_k$ ways of performing each respective operation.

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The total number of possibilities is given by

$$n_1 \times n_2 \times \cdots \times n_k$$

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How many distinct ice cream orders are possible?

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- Three types of cone (cups are not available)
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How many distinct ice cream orders are possible?

$$3 \times 10 \times (4 + 1) = \boxed{150}$$

(Note: you can order an ice cream without any topping, hence 5 topping possibilities)

Example 4: License plate numbers

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If in 2021, the state releases a new series of plate numbers that must now end with a letter in addition to 5 preceding digits, how many distinct plates can be manufactured?

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$$10 \times 10 \times 10 \times 10 \times 10 = 10^5$$

If in 2021, the state releases a new series of plate numbers that must now end with a letter in addition to 5 preceding digits, how many distinct plates can be manufactured? Answer: $26 \times 10^5 = 2,600,000$

Permutations

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- $4! = 4 \times 3 \times 2 \times 1 =$

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- $5! =$

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$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1 \quad (13)$$

Examples of factorials

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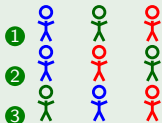
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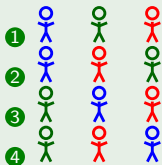
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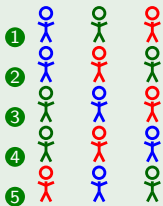
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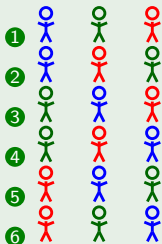
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Permutations with repetition

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Without repetition, the number of permutations of n objects is $n!$.

However, if there are n_1 identical (or repeated) items of type 1, n_2 identical items of type 2, $\dots$, and n_k identical items of type k , then the number of permutations is given by

$$\frac{n!}{n_1! n_2! \times \dots \times n_k!} \quad (15)$$

How many ways can the letters in the word “MASS” be arranged?

The letter “S” is repeated. Thus, the number of permutations are:

$$\frac{4!}{2!} = 4 \times 3 = 12$$

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With combinations, the order is not important (in contrast to permutations).

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You can form 56 different teams.

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